PUBLIC WHITE PAPER

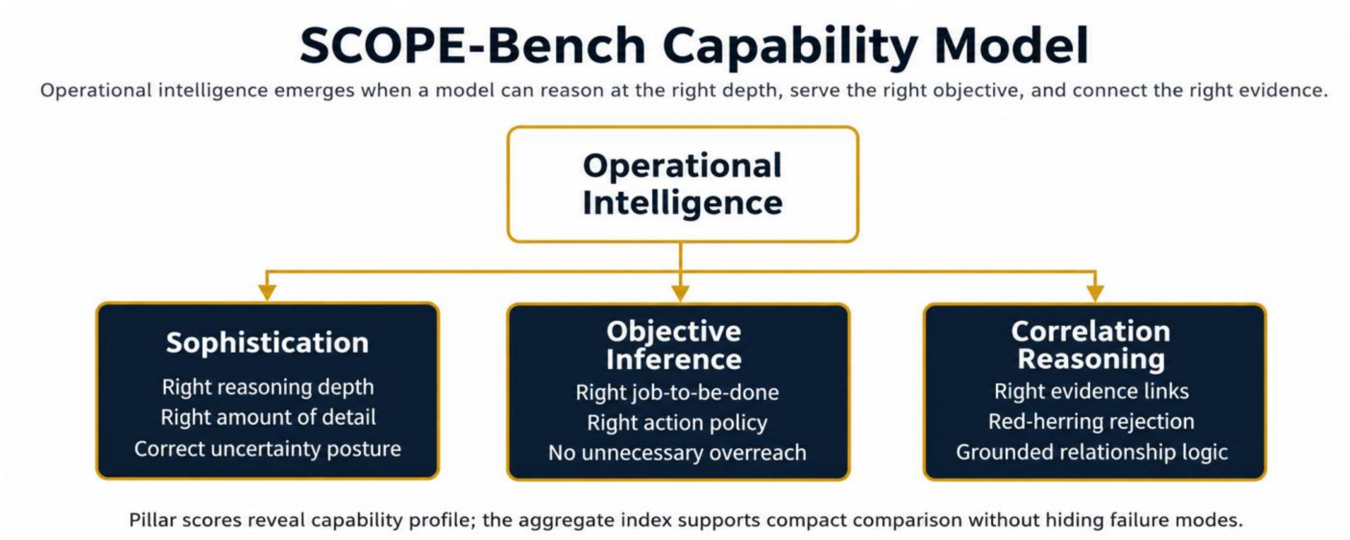
Coeus-SCOPE-Bench

A Benchmark for Measuring Operational Intelligence in Large Language Models

Sophistication | Objective Inference | Correlation Reasoning

Coeus Institute
June 2026

Repository: github.com/CoeusInstitute/SCOPE-Benchmark | Interactable Result Data: https://coeus.institute



Abstract

Large language model benchmarks increasingly influence research priorities, procurement decisions, product claims, and enterprise deployment choices. Yet many widely cited benchmarks emphasize static knowledge, explicit instruction compliance, or narrow task performance while leaving a critical operational layer under-measured: whether a model can calibrate the depth of its reasoning, infer the real objective behind an imperfect or nuanced request, and discover justified relationships across fragmented evidence without over-correlating noise. SCOPE-Bench is a Coeus Institute benchmark framework designed to evaluate that missing layer. It measures three pillars: Sophistication, Objective Inference, and Correlation Reasoning. The benchmark uses procedurally varied item families, hidden-instance generation, inverse cases, objective verification where feasible, shortcut-baseline testing, and diagnostic pillar reporting. Its purpose is not to replace broad academic benchmarks, but to complement them with an operational capability assay aimed at executive assistants, analytical workflows, enterprise intelligence systems, agentic engineering, investigative drafting, and decision-support systems. This public paper presents the benchmark rationale, pillar definitions, test families, methodology, differentiation, limitations, and intended reporting model while withholding exact scoring formulas and other implementation details required to preserve benchmark integrity.

This paper describes SCOPE-Bench at a publishable methodology level. It discloses the public reporting contract, including the official aggregate ranking formula, while intentionally excluding proprietary scoring weights, hidden seed policy mechanics, verifier thresholds, item-generation parameters, and implementation-specific safeguards that would make the benchmark easier to game.

Executive Summary

The industry gap: Many LLM benchmarks still reward static answer accuracy, direct instruction compliance, or narrow task performance. These are useful, but they do not fully test whether a model behaves well inside real operational workflows.

SCOPE-Bench focus: SCOPE-Bench evaluates three capabilities that routinely determine whether an LLM is useful in enterprise and analytical settings: calibrated reasoning, objective recovery, and evidence correlation.

Why the pillars matter: Professional users rarely fail because a model lacks one isolated fact. They fail when a model overthinks or underthinks, answers the wrong implicit problem, misses a relationship across documents, or accepts a false correlation.

How it differs: SCOPE-Bench is designed around procedurally instantiated item families, private live instances, inverse traps, objective verifiers, and shortcut-baseline resistance rather than a single static set of public prompts.

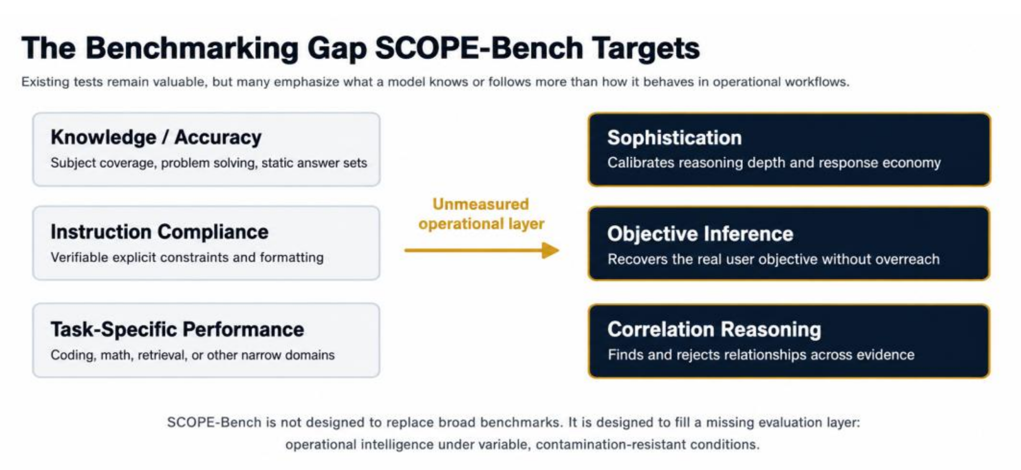


Figure 1. SCOPE-Bench targets an under-measured operational evaluation layer: calibrated reasoning, objective recovery, and correlation reasoning.

Contents

1. The Benchmarking Problem

Evaluation has become one of the central bottlenecks in understanding large language model capability. Benchmark tables influence model selection, public perception, investor narratives, procurement choices, and the internal confidence organizations place in AI systems. However, as models become stronger and benchmarks become more visible, a benchmark score can begin to measure exposure to a known artifact as much as it measures the underlying capability.

This does not mean that existing benchmarks are unimportant. MMLU (Massive Multitask Language Understanding) helped establish broad multitask knowledge evaluation across academic and professional subjects; BIG-bench expanded the field toward diverse and difficult behavioral probes; HELM emphasized multi-metric transparency; IFEval demonstrated the value of objective, verifiable instruction-following checks; SWE-bench moved software evaluation toward real issue resolution; and LiveBench advanced contamination-limited evaluation through regularly refreshed, objectively scored questions. Each of these efforts contributed a necessary layer of the evaluation landscape.

The problem is that the industry increasingly uses benchmark aggregates as if they directly describe operational fitness. In deployed environments, a model does not merely answer multiple-choice questions, satisfy formatting constraints, or patch an isolated repository issue. It must interact with incomplete instructions, ambiguous objectives, mixed evidence, changing context, organizational data, hidden constraints, and human workflow expectations. SCOPE-Bench was created to test that missing operational layer.

2. Coeus Institute R&D Context

Coeus Institute designs and develops autonomous intelligence systems that transform large, messy, fragmented, and often qualitative data environments into structured, decision-ready intelligence. Its work sits at the intersection of AI orchestration, data automation, enterprise intelligence, strategic analysis, and operational decision support. In practical terms, Coeus builds systems that ingest diverse information, structure it, identify meaningful patterns, surface risk and opportunity signals, and support decision-makers with context-aware recommendations.

That R&D context shaped SCOPE-Bench. Coeus systems are not built around a single clean prompt-response pattern. They are built for environments where data arrives from documents, logs, codebases, meeting records, emails, open-source information, operational reports, customer records, and domain-specific knowledge bases. In those environments, model selection cannot be reduced to raw knowledge accuracy. The deciding question is whether a model can behave intelligently inside the workflow.

SCOPE-Bench therefore emerged from a deployment-driven need: a way to compare models not only by what they know, but by whether they can reason at the right level, understand the real objective, and connect evidence across distributed information landscapes.

3. The Major Hole in Current LLM Benchmarking

The major hole SCOPE-Bench targets is the absence of a rigorous, operational test for model behavior under realistic ambiguity and evidence fragmentation. Current benchmark categories often measure discrete competence: knowledge recall, mathematical reasoning, coding performance, summarization, instruction adherence, safety, tool use, or domain-specific accuracy. These remain valuable, but they frequently miss three recurring failure modes that matter in production.

First, models often miscalibrate reasoning depth. They under-process subtle tasks or over-process simple ones. Second, models often solve the stated request rather than the actual objective, especially when the user gives an incomplete, shorthand, or mis-specified instruction. Third, models often retrieve evidence without correctly correlating it; they miss multi-hop relationships or accept plausible but false links.

These failures are not cosmetic. They create wasted executive time, weak analysis, brittle agentic behavior, false operational narratives, incorrect technical diagnoses, and poor decision support. SCOPE-Bench is designed around those failure modes.

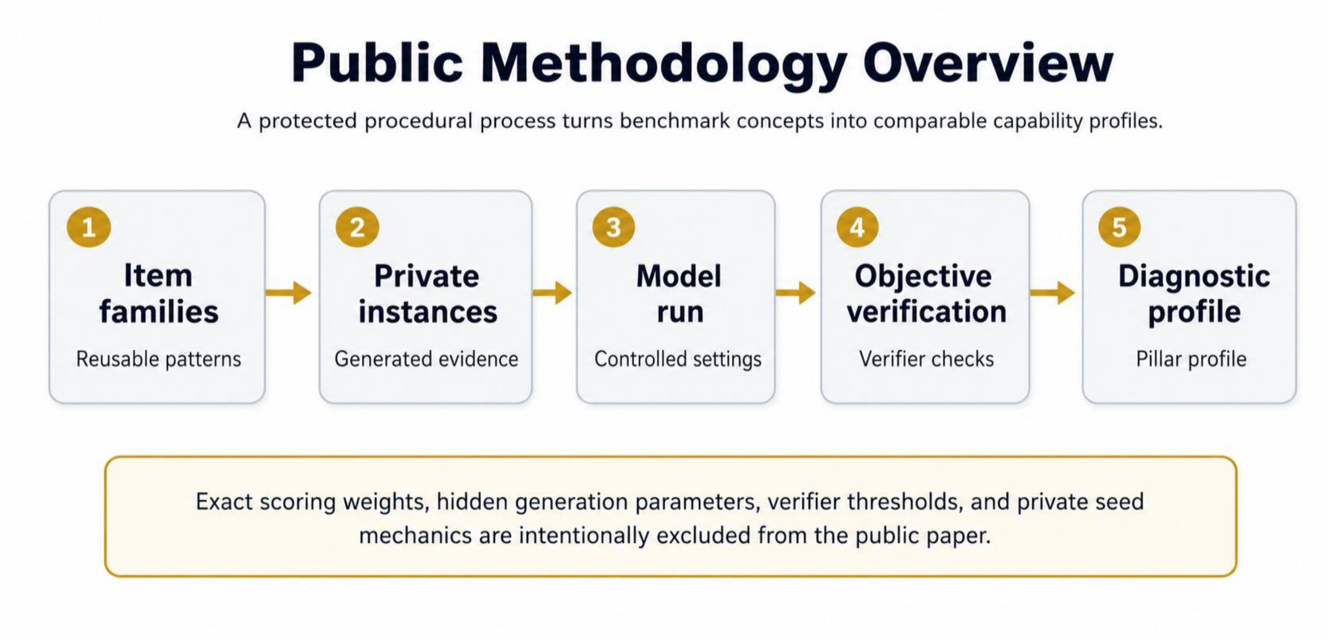


Figure 2. SCOPE-Bench uses procedural item families and objective verification to generate diagnostic capability profiles while protecting benchmark integrity.

4. SCOPE-Bench Framework Overview

SCOPE-Bench is a procedurally instantiated benchmark framework for evaluating operational intelligence in large language models. The benchmark is organized around three pillars: Sophistication, Objective Inference, and Correlation Reasoning. Together, these pillars ask whether a model can choose the right depth of reasoning, serve the correct objective, and determine which relationships across evidence are real.

The framework is intentionally not defined by a single static public dataset. Public examples can demonstrate the benchmark design, but official evaluations should rely on protected live instances generated from item families. This makes the benchmark more resistant to contamination, memorization, prompt overfitting, and shallow leaderboard optimization.

Each pillar produces diagnostic information. A model can perform well on correlation tasks while failing objective inference; it can infer user goals while over-correlating evidence; it can produce nuanced prose while missing hidden constraints. SCOPE-Bench is designed to show those differences rather than collapse them into an opaque one-dimensional claim.

A compact aggregate index remains useful for model comparison, but it is not the whole benchmark. The benchmark's primary value is the capability profile: where a model is strong, where it is brittle, and what kind of operational deployment it can responsibly support.

The three pillars were selected because they satisfy five design criteria:

Criterion	Benchmark design meaning
Operational relevance	The capability must affect real enterprise, engineering, analytical, or decision-support workflows.
Under-measurement	The capability must not already be sufficiently captured by standard accuracy, instruction-following, or code-generation scores.
Observable failure modes	The capability must produce recurring errors that can be identified and analyzed.
Verifiability	The benchmark must reduce reliance on subjective judging wherever possible.
Contamination resistance	The capability must be testable through varied instances rather than fixed answer artifacts.

5. Pillar One: Sophistication

A sophisticated model does not always think more. It thinks enough. In enterprise use, a model that turns a simple instruction into a long, caveated essay is operationally expensive. A model that treats a subtle legal, investigative, financial, or engineering request as a simple lookup is risky. SCOPE-Bench tests both errors.

Sophistication is the model's ability to apply the minimum sufficient reasoning depth, abstraction level, uncertainty handling, and synthesis required by the task. It is not verbosity. It is calibrated cognitive economy.

This pillar matters for executive assistants, legal and investigative drafting, technical support, strategic analysis, and agentic systems that must decide whether to answer directly, reason deeper, summarize, escalate, or preserve uncertainty.

Public test families

Test family	What it evaluates
Depth calibration tests	These items evaluate whether the model recognizes when a task contains hidden constraints, non-obvious dependencies, or a need for deeper synthesis.
Simplicity-trap tests	These items evaluate whether the model can avoid overcomplicating a direct request, adding unnecessary caveats, or substituting generic analysis for the requested answer.
Discrimination tests	These items present superficially similar cases that differ by a decisive condition. The model must identify the relevant distinction rather than pattern-match loosely.
Uncertainty calibration tests	These items evaluate whether the model expresses uncertainty when evidence is incomplete and avoids unsupported certainty when the record is underdetermined.
Response economy tests	These items examine whether the model can deliver a complete answer at an appropriate level of detail rather than hiding weak reasoning behind excessive output.

6. Pillar Two: Objective Inference

Instruction following asks whether the model did what the user said. Objective inference asks whether the model solved what the user was trying to accomplish. Real users rarely specify every constraint or objective cleanly. They use shorthand, point to prior context, name the wrong mechanism, or ask for an intermediate artifact when the actual need is strategic. Objective Inference is the model's ability to recover and serve the user's real objective when the stated request is incomplete, indirect, ambiguous, constrained by context, or mis-specified.

This pillar is intentionally balanced against overreach. The goal is not aggressive mind-reading. A high-performing model must know when to infer, when to ask, and when to execute literally.

Public test families

Test family	What it evaluates
Inferable latent-objective tests	The user provides enough cues to infer the real job-to-be-done. The model should infer and act rather than ask unnecessary questions.
Ambiguity-triage tests	The user's request is genuinely underspecified. The model should ask a targeted clarification instead of inventing intent.
Literal-objective tests	The user means exactly what was stated. The model should execute literally and not hallucinate a hidden agenda.
Mis-specified-request tests	The user asks for a means that will not achieve the apparent goal. The model should correct the framing and move toward the real objective.
Implicit-constraint tests	The model must preserve obvious contextual constraints, such as audience, confidentiality, format, scope, or deployment setting.

7. Pillar Three: Correlation Reasoning

Many high-value analytical tasks are not solved by retrieval alone. The relevant signal may be distributed across a meeting note, a metric anomaly, a code change, a support ticket, and a policy memo. The model must determine which pieces of evidence belong together and which tempting links should be rejected.

This pillar matters for enterprise intelligence, OSINT (Open Source Intelligence)-style analysis, incident review, codebase reasoning, compliance workflows, customer operations, and executive decision support. In these environments, a false relationship can be more dangerous than a missed one because it can produce a persuasive but wrong operational narrative.

Correlation Reasoning is the model's ability to discover, justify, and reject relationships across distributed heterogeneous evidence, including qualitative documents, quantitative records, logs, code-like files, timestamps, aliases, and operational artifacts.

Public test families

Test family	What it evaluates
Multi-document relationship tests	The model must connect evidence across documents rather than rely on one isolated passage.
Evidence-grounded graph tests	The model must identify the relevant relationship and cite or otherwise ground the supporting evidence.
Red-herring rejection tests	The model must reject plausible but false relationships, repeated-entity traps, or surface-level correlations.
Temporal and alias tests	The model must resolve ordering, lag, naming variation, entity aliases, and causal sequence.
Code-to-metric tests	The model must connect implementation changes, configuration details, logs, incidents, or observed metrics without turning correlation into unsupported causation.

8. Public Methodology

SCOPE-Bench is designed as a procedural benchmark rather than a fixed public test artifact. At the public level, this means the benchmark is organized around reusable item families that can generate varied instances. Official items should be created from protected parameters and private seeds so that models cannot simply memorize a known prompt set.

The benchmark prioritizes objective or semi-objective verification. Where a task can be checked through answer keys, required and forbidden evidence, structured output validation, or relationship matching, those mechanisms are preferred over open-ended model-as-judge evaluation. Human review and model-based judging may still be useful during item development, but they are not the desired foundation for official scoring.

Each pillar includes inverse cases. These are tasks designed to catch shallow strategies that often inflate benchmark performance. For example, always asking a clarification question should fail when the objective is inferable; always answering directly should fail when the request is genuinely ambiguous; always producing a long answer should fail on simplicity traps; always correlating repeated entities should fail on red-herring items.

The benchmark also uses shortcut baselines during validation. If an item family can be solved well by a simple generic policy, such as always answering directly, always refusing ambiguity, or always selecting the most semantically similar evidence, then that item family is not mature enough for official evaluation.

Benchmark lifecycle

Stage	Public description
Design	Define the capability, test family, expected behavior, distractors, and failure modes.
Generate	Create procedurally varied instances with protected details and known ground truth.
Execute	Run models under controlled settings with clearly labeled tool and context policies.
Verify	Score outputs through objective checks wherever feasible.
Diagnose	Report pillar-level scores, failure modes, efficiency signals, and reliability indicators.
Audit	Test against shortcut baselines and revise item families that are too easily gamed.

9. Reporting Model and Benchmark Integrity

SCOPE-Bench is intended to publish both a compact comparative result and a richer diagnostic profile. The aggregate result supports high level comparison, while the pillar and failure-mode diagnostics explain why a model received that result and where it is likely to be operationally brittle.

The public methodology discloses the reporting contract needed to interpret results, including the official aggregate ranking formula, but does not disclose proprietary item weights, verifier thresholds, verifier internals, hidden ground-truth schemas, item-generation parameters, or live seed policies. This is not an attempt to avoid scrutiny. It is a benchmark integrity requirement. Over disclosure would make it easier for models, prompts, or wrappers to optimize against the benchmark's surface mechanics instead of improving the underlying capability.

Published results should therefore disclose enough for meaningful interpretation: benchmark version, model version, run configuration, tool mode, item counts by pillar, public examples, failure categories, reliability indicators, and confidence intervals where available. They should not disclose the protected mechanics that would collapse the benchmark into a static target.

10. Real-World Use-Case Value

The three pillars were selected because they map to practical deployment risks. In executive-support systems, the model must know when the user needs a direct answer and when they need synthesis. In agentic engineering, the model must decide whether a request is a surface symptom or a deeper system problem. In enterprise intelligence, the model must correlate scattered evidence without accepting false positives. In investigative or compliance workflows, the model must preserve constraints and avoid unsupported certainty.

SCOPE-Bench therefore helps answer a different model-selection question: not simply "Which model is generally smarter?" but "Which model behaves reliably inside the operational pattern we intend to deploy?" A model with high Objective Inference but weak Correlation Reasoning may be strong in user-facing assistance but weak for evidence synthesis. A model with strong Correlation Reasoning but poor Sophistication may find links while overproducing unwieldy analysis. A model with balanced pillar scores may be better suited for autonomous or semi-autonomous workflows.

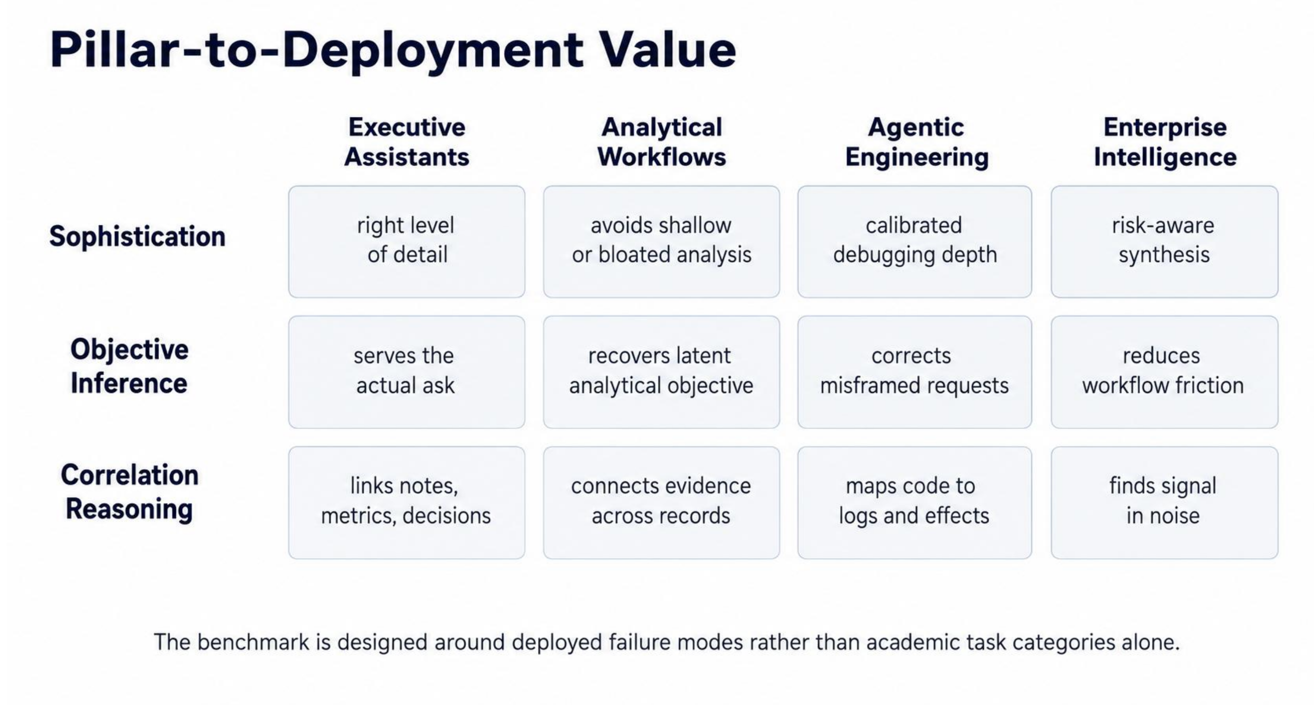


Figure 3. The three pillars map directly to deployed enterprise and analytical failure modes.

11. Positioning Against Existing Benchmarks

SCOPE-Bench is designed to complement existing evaluation frameworks. Broad knowledge benchmarks are useful for subject coverage. Instruction-following benchmarks are useful for explicit constraint compliance. Software benchmarks are useful for codebase repair. Dynamic benchmarks are useful for reducing contamination from static public artifacts. SCOPE-Bench occupies a different layer: operational intelligence under ambiguity, evidence fragmentation, and workflow pressure.

The benchmark's distinct contribution is not that it is the only benchmark to use objective scoring or contamination-resistant practices. Rather, its contribution is the combination of three operational constructs, inverse item families, procedural instantiation, shortcut-baseline validation, and diagnostic reporting aimed at real enterprise deployment decisions.

Evaluation layer	Representative focus	Strength	SCOPE-Bench distinction
Broad knowledge benchmarks	Academic/professional subject accuracy	Useful coverage signal	Adds operational behavior and procedural variation
Instruction-following benchmarks	Explicit verifiable constraints	Objective compliance checks	Tests latent objectives, ambiguity triage, and mis-specified requests
Software/task benchmarks	Real task execution in specific domains	High ecological validity for target domain	Tests cross-evidence correlation beyond patching alone
Dynamic benchmarks	Refreshing questions and reducing contamination	Improves integrity against static memorization	Combines contamination resistance with operational capability pillars

12. Limitations and Responsible Use

SCOPE-Bench does not claim to measure intelligence, consciousness, general judgment, safety, fairness, truthfulness, or regulatory fitness in the abstract. It measures operationally observable subsets of three target constructs. A high score should not be treated as a universal deployment license.

Synthetic and semi-synthetic corpora allow known ground truth, but they may not capture every property of real organizational data. Real-data tracks can improve ecological validity, but they require permissioned, de-identified, or otherwise authorized data with carefully established ground truth. Both approaches have tradeoffs.

Objective verification improves reproducibility but can be rigid. Some valid alternative answers may be missed if verifiers are poorly designed. SCOPE-Bench should therefore maintain rigorous item review, positive and negative controls, shortcut baselines, and periodic audits.

Finally, no benchmark can permanently prevent gaming. SCOPE-Bench reduces dependence on static artifacts, but benchmark integrity requires ongoing renewal, hidden live instances, external scrutiny, and a clear separation between public samples and official evaluation material.

13. Conclusion

SCOPE-Bench addresses a practical gap in LLM evaluation: the need to measure operational intelligence rather than only static accuracy, explicit instruction compliance, or narrow task performance. Its three pillars - Sophistication, Objective Inference, and Correlation Reasoning - reflect capabilities that repeatedly determine whether a model can function inside real enterprise, analytical, engineering, and decision-support workflows.

The benchmark is designed to be contamination-resistant, shortcut-resistant, and diagnostically useful. It uses procedural item families, protected live instances, inverse cases, objective verification where feasible, and pillar-level reporting. Its public methodology is transparent about the constructs and design principles while protecting the exact mechanics required to preserve benchmark validity.

As LLMs move from chat interfaces into autonomous and semi-autonomous systems, evaluation must evolve from “Can the model answer this known task?” to “Can the model behave intelligently inside the operational system?” SCOPE-Bench is Coeus Institute’s contribution to that next layer of evaluation.

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Appendix A. SCOPE-Bench Results

Evaluation date: June 30, 2026

Run ID: 9ff7a720-826a-4d34-9fbf-ca21b7a70b39

Run name: official-current-2026-06-30

This appendix reports the live completed official-current-2026-06-30 evaluation from the SCOPE-Bench database. The run evaluated 28 models across the active 21-item official-current public calibration suite: 6 Sophistication items, 7 Objective Inference items, and 8 Correlation Reasoning items. The database contains 588 saved model responses for this run, matching 28 models x 21 items.

Official scoring contract: item scores use scope-score-v0.8.2, aggregate reports use scope-combo-v0.8.4, and the official model ranking value is combo_score = 100 * ((P_S + P_I + P_C) / 3). Token use, duration, latency, cost, reliability, balance, SCOPE Base, and relative efficiency are diagnostics only; they do not modify item scores, pillar scores, or combo_score.

The run reflects the current benchmark pipeline: dashboard-submitted run creation, Supabase Edge Function execution through OpenRouter, raw response persistence in model_invocations, deterministic Python scoring into item_scores, aggregate_reports, and typed model_run_results projection rows for dashboard and paper reporting.

A.1 Evaluation Composition

Evaluation component	Value
Run name	official-current-2026-06-30
Run ID	9ff7a720-826a-4d34-9fbf-ca21b7a70b39
Duration	93m 8s
Models evaluated	28
Benchmark items per model	21
Item score formula	scope-score-v0.8.2
Aggregate formula	scope-combo-v0.8.4
Official ranking metric	combo_score
Top-level aggregate combo	77.965071
Top-level pass / scored rows	367/588
Incorrect rows	210
Provider-error rows	11

A.2 Official Model Ranking

Table A.2 ranks all 28 models by combo_score. Pillar scores are shown on the same 0-100 scale for diagnostic interpretation.

Rank	Model	Combo	Soph.	Obj.	Corr.	Pass/Items
1	minimax/minimax-m3	93.412	96.53	88.52	95.19	18/21
2	anthropic/claude-opus-4.8	88.370	76.94	90.57	97.59	17/21
3	anthropic/claude-opus-4.7	87.768	79.44	87.47	96.39	16/21
4	qwen/qwen3.6-flash	87.165	85.35	79.76	96.39	15/21
5	google/gemini-3.1-flash-lite	86.801	82.43	81.33	96.65	15/21
6	openai/gpt-5.5	86.384	84.86	88.13	86.16	17/21
7	google/gemma-4-31b-it	84.944	77.43	81.36	96.04	14/21
8	google/gemini-3.5-flash	84.912	77.99	80.36	96.39	15/21
9	meta-llama/llama-4-maverick	83.876	76.32	82.98	92.33	14/21
10	openai/gpt-5.4-mini	83.286	78.61	85.69	85.56	16/21
11	x-ai/grok-build-0.1	83.022	76.11	76.57	96.39	14/21
12	deepseek/deepseek-v4-flash	82.880	69.72	82.53	96.39	14/21
13	qwen/qwen3.7-plus	82.349	82.85	79.25	84.95	14/21
14	x-ai/grok-4.20-multi-agent	81.128	74.38	72.62	96.39	12/21
15	moonshotai/kimi-k2.7-code	80.943	72.43	86.33	84.07	14/21
16	nvidia/nemotron-3-ultra-550b-a55b	80.567	70.28	87.35	84.07	14/21
17	deepseek/deepseek-v4-pro	79.841	64.79	78.34	96.39	13/21
18	x-ai/grok-4.3	79.509	70.00	72.14	96.39	13/21
19	mistralai/mistral-small-2603	78.666	70.28	70.96	94.76	12/21
20	z-ai/glm-5.2	76.083	83.89	84.04	60.32	14/21
21	xiaomi/mimo-v2.5-pro	74.394	62.50	87.44	73.24	12/21
22	minimax/minimax-m2.7	74.008	83.89	89.85	48.29	14/21
23	meta-llama/llama-4-scout	70.083	46.04	74.34	89.87	8/21
24	openai/gpt-oss-120b	67.965	54.44	75.33	74.12	9/21
25	mistralai/mistral-medium-3-5	66.888	64.79	74.67	61.20	11/21
26	moonshotai/kimi-k2.6	64.077	76.32	73.04	42.87	10/21
27	tencent/hy3-preview	50.257	66.88	59.04	24.86	5/21
28	qwen/qwen3.7-max	43.442	75.69	28.70	25.93	7/21

A.3 Operational Details

Operational fields provide deployment context. They are not substitutes for the official benchmark score and are not used as score modifiers.

Model	Pass Rate	Failures	Scored Tokens	Total Tokens	Estimated Cost	Duration
minimax/minimax-m3	86%	3	21,791	21,604	$0.01777	3m 1s
anthropic/claude-opus-4.8	81%	4	13,707	21,644	$0.34602	2m 55s
anthropic/claude-opus-4.7	76%	5	5,603	15,282	$0.18487	1m 35s
qwen/qwen3.6-flash	71%	6	62,962	39,494	$0.038237	3m 46s
google/gemini-3.1-flash-lite	71%	6	2,903	9,391	$0.0059765	0m 36s
openai/gpt-5.5	81%	4	20,870	17,844	$0.385095	3m 40s
google/gemma-4-31b-it	67%	7	3,204	13,526	$0.00376627	1m 28s
google/gemini-3.5-flash	71%	6	36,947	26,704	$0.191676	1m 50s
meta-llama/llama-4-maverick	67%	7	3,902	10,008	$0.00580294	0m 49s
openai/gpt-5.4-mini	76%	5	19,338	17,018	$0.054047	2m 6s
x-ai/grok-build-0.1	67%	7	74,378	46,595	$0.083437	4m 48s
deepseek/deepseek-v4-flash	67%	7	25,877	20,205	$0.00336648	2m 44s
qwen/qwen3.7-plus	67%	7	34,613	24,582	$0.025682	7m 14s
x-ai/grok-4.20-multi-agent	57%	9	956,106	1,793,488	$1.932836	4m 50s
moonshotai/kimi-k2.7-code	67%	7	26,311	20,343	$0.067795	4m 1s
nvidia/nemotron-3-ultra-550b-a55b	67%	7	30,555	21,961	$0.048671	4m 57s
deepseek/deepseek-v4-pro	62%	8	29,723	22,283	$0.060397	4m 29s
x-ai/grok-4.3	62%	8	45,037	31,700	$0.06677	3m 36s
mistralai/mistral-small-2603	57%	9	26,562	19,644	$0.00877257	1m 29s
z-ai/glm-5.2	67%	7	37,299	24,063	$0.128617	1m 50s
xiaomi/mimo-v2.5-pro	57%	9	44,846	29,755	$0.074572	7m 51s
minimax/minimax-m2.7	67%	7	41,340	26,160	$0.043839	1m 20s
meta-llama/llama-4-scout	38%	13	3,560	10,625	$0.00425825	0m 34s
openai/gpt-oss-120b	43%	12	44,150	30,783	$0.01929	0m 48s
mistralai/mistral-medium-3-5	52%	10	54,090	32,210	$0.203061	5m 8s
moonshotai/kimi-k2.6	48%	11	57,538	35,518	$0.107466	2m 37s
tencent/hy3-preview	24%	16	62,233	37,995	$0.00868773	4m 42s
qwen/qwen3.7-max	33%	14	13,671	9,759	$0.030646	3m 44s

A.4 Relative Efficiency Diagnostics

Relative token and cost efficiency are secondary diagnostics normalized to the best model in this selected run. A value of 100 means best-in-run score-per-token or score-per-dollar, not perfect benchmark quality.

Model	Rel. Token Eff.	Rel. Cost Eff.	Combo / 1k Scored Tokens
minimax/minimax-m3	14.337	21.352	4.287
anthropic/claude-opus-4.8	21.562	1.037	6.447
anthropic/claude-opus-4.7	52.389	1.928	15.664
qwen/qwen3.6-flash	4.630	9.259	1.384
google/gemini-3.1-flash-lite	100.000	58.993	29.900
openai/gpt-5.5	13.843	0.911	4.139
google/gemma-4-31b-it	88.667	91.610	26.512
google/gemini-3.5-flash	7.686	1.799	2.298
meta-llama/llama-4-maverick	71.891	58.710	21.496
openai/gpt-5.4-mini	14.404	6.259	4.307
x-ai/grok-build-0.1	3.733	4.042	1.116
deepseek/deepseek-v4-flash	10.712	100.000	3.203
qwen/qwen3.7-plus	7.957	13.024	2.379
x-ai/grok-4.20-multi-agent	0.284	0.170	0.085
moonshotai/kimi-k2.7-code	10.289	4.850	3.076
nvidia/nemotron-3-ultra-550b-a55b	8.819	6.724	2.637
deepseek/deepseek-v4-pro	8.984	5.370	2.686
x-ai/grok-4.3	5.904	4.837	1.765
mistralai/mistral-small-2603	9.905	36.424	2.962
z-ai/glm-5.2	6.822	2.403	2.040
xiaomi/mimo-v2.5-pro	5.548	4.052	1.659
minimax/minimax-m2.7	5.987	6.857	1.790
meta-llama/llama-4-scout	65.839	66.851	19.686
openai/gpt-oss-120b	5.148	14.311	1.539
mistralai/mistral-medium-3-5	4.136	1.338	1.237
moonshotai/kimi-k2.6	3.725	2.422	1.114
tencent/hy3-preview	2.701	23.497	0.808
qwen/qwen3.7-max	10.627	5.758	3.178

A.5 Pillar Leaders

A.5.1 Sophistication

Rank	Model	Sophistication	Combo
1	minimax/minimax-m3	96.53	93.412
2	qwen/qwen3.6-flash	85.35	87.165
3	openai/gpt-5.5	84.86	86.384
4	z-ai/glm-5.2	83.89	76.083
5	minimax/minimax-m2.7	83.89	74.008
6	qwen/qwen3.7-plus	82.85	82.349
7	google/gemini-3.1-flash-lite	82.43	86.801
8	anthropic/claude-opus-4.7	79.44	87.768

A.5.2 Objective Inference

Rank	Model	Objective Inference	Combo
1	anthropic/claude-opus-4.8	90.57	88.370
2	minimax/minimax-m2.7	89.85	74.008
3	minimax/minimax-m3	88.52	93.412
4	openai/gpt-5.5	88.13	86.384
5	anthropic/claude-opus-4.7	87.47	87.768
6	xiaomi/mimo-v2.5-pro	87.44	74.394
7	nvidia/nemotron-3-ultra-550b-a55b	87.35	80.567
8	moonshotai/kimi-k2.7-code	86.33	80.943

A.5.3 Correlation Reasoning

Rank	Model	Correlation Reasoning	Combo
1	anthropic/claude-opus-4.8	97.59	88.370
2	google/gemini-3.1-flash-lite	96.65	86.801
3	anthropic/claude-opus-4.7	96.39	87.768
4	qwen/qwen3.6-flash	96.39	87.165
5	google/gemini-3.5-flash	96.39	84.912
6	x-ai/grok-build-0.1	96.39	83.022
7	deepseek/deepseek-v4-flash	96.39	82.880
8	x-ai/grok-4.20-multi-agent	96.39	81.128

A.6 Results Summary

The live June 30 run produced an aggregate combo_score of 77.965 across 588 scored rows, with a strict pass rate of 62%. The highest-ranked model was minimax/minimax-m3, with combo_score 93.412, Sophistication 96.53, Objective Inference 88.52, and Correlation Reasoning 95.19.

The 28-model panel shows meaningful spread across the three SCOPE-Bench pillars. The top overall models did not all win the same diagnostic dimensions: minimax/minimax-m3 led Sophistication, anthropic/claude-opus-4.8 led Objective Inference, and anthropic/claude-opus-4.8 led Correlation Reasoning. This reinforces the benchmark design goal of preserving pillar-level profiles rather than reducing every deployment decision to a single opaque score.

Efficiency diagnostics also varied substantially. google/gemini-3.1-flash-lite had the best relative token efficiency, while deepseek/deepseek-v4-flash had the best relative cost efficiency. These fields support deployment analysis but remain separate from official model capability scoring.

A.7 Configuration Notes

Configuration field	Value
Evaluation date	June 30, 2026
Benchmark version	official-current
Item set version	official-current
Run source	Supabase benchmark_runs, aggregate_reports, model_run_results, and model_invocations
Run ID	9ff7a720-826a-4d34-9fbf-ca21b7a70b39
Item score formula	scope-score-v0.8.2
Aggregate formula	scope-combo-v0.8.4
Model provider route	OpenRouter through Supabase Edge Functions
Result persistence	model_invocations, item_scores, aggregate_reports, and model_run_results
Dashboard read path	dashboard-runs Edge Function and React/Vite operator dashboard
Scoring authority	Deterministic Python harness

A.8 Limitations

These results come from one completed official-current run over the public calibration suite. They are suitable for paper-facing methodology and operational comparison, but they should not be treated as a final private benchmark leaderboard. Future official campaigns should use private/generated item sets, larger repeated-run analysis where appropriate, version-pinned run configurations, and confidence intervals for publication-grade rankings.